Exercise 2.1

Question 1 - Compare how you added the above data and how it differs from INSERT records in a relational database

In relational databases, data is stored in tables consisting of rows and columns. Before inserting records, the table structure must be explicitly defined using SQL '*CREATE TABLE*' statements. In contrast, MongoDB stores data in collections as documents (JSON-like objects).

```
> db.dept.insert( { _id: 30, deptno: 30, dname: 'SALES', loc: 'CHICAGO'})
< DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
< {
    acknowledged: true,
    insertedIds: {
      '0': 30
    }
  }
```

Figure 1: Inserting data in MongoDB

Similarly, in relational databases, a primary key must be explicitly defined whereas in MongoDB *_id* acts as the primary key.

Question 2 - Add Department 30, which has the following key/values: id: 30, deptno: 30, dname: SALES and loc: CHICAGO

```
> use dept
< switched to db dept
> db.dept.insert( { _id: 30, deptno: 30, dname: 'SALES', loc: 'CHICAGO'})
< DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
< {
    acknowledged: true,
    insertedIds: {
      '0': 30
    }
  }
```

```
> db.dept.find()
< {
    _id: 30,
    deptno: 30,
    dname: 'SALES',
    loc: 'CHICAGO'
  }
Enterprise dept>
```

Exercise 2.2

<u>Question 1 – Add the employees for Department 30</u>

```
> db.emp.insertOne(
    {
      empno: 7499,
      ename: 'ALLEN',
      job: 'SALESMAN',
      mgr: 7698,
      hiredate: new Date('1995-02-20'),
      sal: 1600,
      comm: 300,
      deptno: 30
    })
< {
    acknowledged: true,
    insertedId: ObjectId('69bcad2b9204a497d5736b11')
  }
```

```
> db.emp.find()
< {
    _id: ObjectId('69bcad2b9204a497d5736b11'),
    empno: 7499,
    ename: 'ALLEN',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: 1995-02-20T00:00:00.000Z,
    sal: 1600,
    comm: 300,
    deptno: 30
  }
```

```
> db.emp.insertMany([
    {
      empno: 7698,
      ename: 'BLAKE',
      job: 'MANAGER',
      mgr: 7839,
      hiredate: new Date('1981-05-01'),
      sal: 2850,
      deptno: 30
    },
    {
      empno: 7900,
      ename: 'JAMES',
      job: 'CLERK',
      mgr: 7698,
      hiredate: new Date('1981-12-03'),
      sal: 950,
      deptno: 30
    },
    {
      empno: 7654,
      ename: 'MARTIN',
      job: 'SALESMAN',
      mgr: 7698,
      hiredate: new Date('1993-10-28'),
      sal: 1250,
      comm: 1400,
      deptno: 30
    },
```

```
  {
    empno: 7844,
    ename: 'TURNER',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: new Date('1981-02-08'),
    sal: 1500,
    comm: 0,
    deptno: 30
  },
  {
    empno: 7521,
    ename: 'WARD',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: new Date('1994-02-22'),
    sal: 1250,
    comm: 500,
    deptno: 30
  },
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69bcae639204a497d5736b12'),
    '1': ObjectId('69bcae639204a497d5736b13'),
    '2': ObjectId('69bcae639204a497d5736b14'),
    '3': ObjectId('69bcae639204a497d5736b15'),
    '4': ObjectId('69bcae639204a497d5736b16')
  }
}
```

```
> db.emp.find()
< {
    _id: ObjectId('69bcad2b9204a497d5736b11'),
    empno: 7499,
    ename: 'ALLEN',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: 1995-02-20T00:00:00.000Z,
    sal: 1600,
    comm: 300,
    deptno: 30
  }
  {
    _id: ObjectId('69bcae639204a497d5736b12'),
    empno: 7698,
    ename: 'BLAKE',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: 1981-05-01T00:00:00.000Z,
    sal: 2850,
    deptno: 30
  }
  {
    _id: ObjectId('69bcae639204a497d5736b13'),
    empno: 7900,
    ename: 'JAMES',
    job: 'CLERK',
    mgr: 7698,
    hiredate: 1981-12-03T00:00:00.000Z,
    sal: 950,
    deptno: 30
  }
```

```
{
  _id: ObjectId('69bcae639204a497d5736b14'),
  empno: 7654,
  ename: 'MARTIN',
  job: 'SALESMAN',
  mgr: 7698,
  hiredate: 1993-10-28T00:00:00.000Z,
  sal: 1250,
  comm: 1400,
  deptno: 30
}
{
  _id: ObjectId('69bcae639204a497d5736b15'),
  empno: 7844,
  ename: 'TURNER',
  job: 'SALESMAN',
  mgr: 7698,
  hiredate: 1981-02-08T00:00:00.000Z,
  sal: 1500,
  comm: 0,
  deptno: 30
}
{
  _id: ObjectId('69bcae639204a497d5736b16'),
  empno: 7521,
  ename: 'WARD',
  job: 'SALESMAN',
  mgr: 7698,
  hiredate: 1994-02-22T00:00:00.000Z,
  sal: 1250,
  comm: 500,
  deptno: 30
}
```

Exercise 2.4

<u>Question 1 – Update the name of the department 40 to: COMPUTING</u>

```
> db.dept.update({'deptno': 40},
                 {$set: {'dname': 'COMPUTING'}} )
< DeprecationWarning: Collection.update() is deprecated. Use updateOne, updateMany, or bulkWr
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
```

```
{
  _id: 40,
  deptno: 40,
  dname: 'COMPUTING',
  loc: 'BOSTON'
}
```

Question 2 – Update the salary of employee number 7788 in department 20 to 3500

```
> db.dept.update({$annd: [{'deptno': 20}, {'empno': 7788} ]},
                 {$set: {'sal': '3500'}} )
✖ ▸ MongoServerError: unknown top level operator: $annd. If you have a field name that starts
Enterprise dept>
```

```
> db.emp.find()
< {
    _id: ObjectId('69bcb3219204a497d5736b17'),
    empno: 7788,
    ename: 'AYAM',
    job: 'MATHEMATICIAN',
    mgr: 7690,
    hiredate: 2024-02-20T00:00:00.000Z,
    sal: 45000,
    comm: 300,
    deptno: 20
  }
Enterprise dept>
```